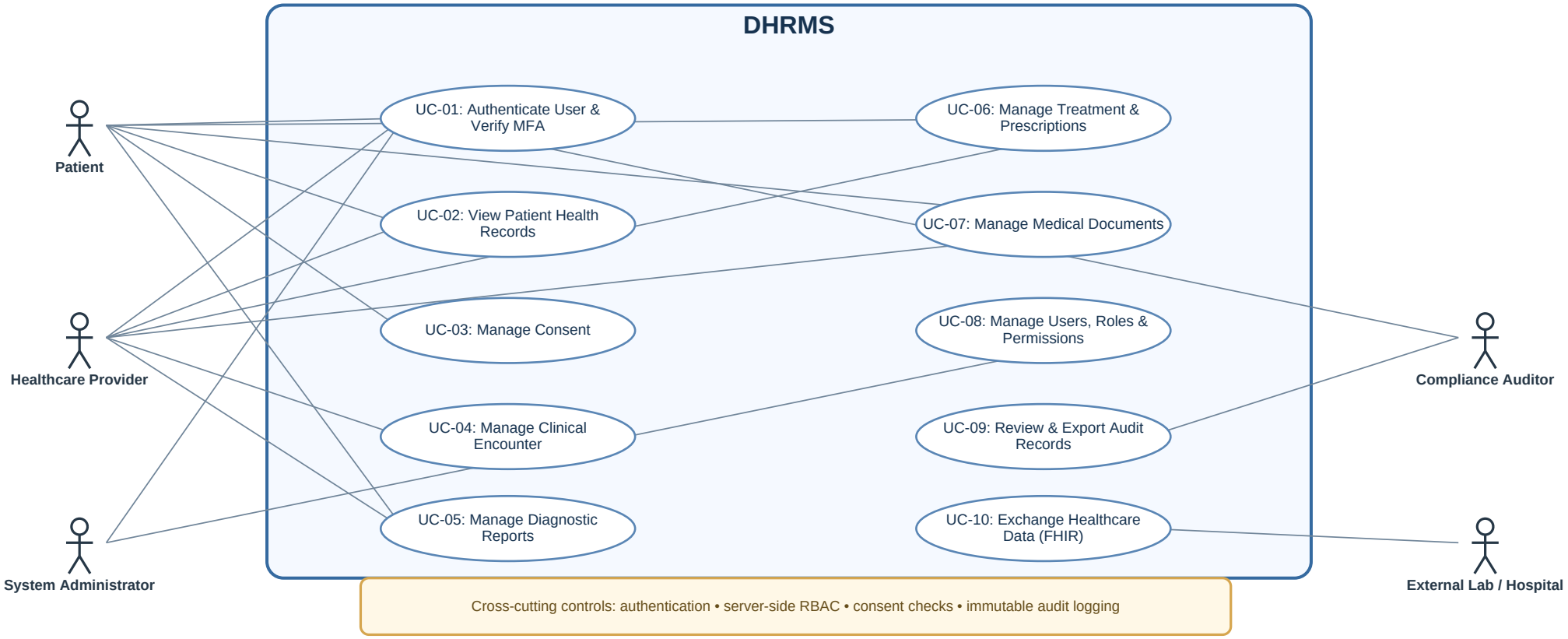
Digital Health Record Management System (DHRMS)

Use Case Diagram, Actor Roles, Preconditions, Postconditions, and Functions | Based on DHRMS SRS v2.0

1. UML Use Case Diagram

The system boundary is DHRMS. Solid lines indicate actor participation in a use case. Security checks apply across protected operations.



2. Actor Roles

Actor	Role / Responsibilities
Patient	Owns health records; views records and prescriptions; grants, reviews, and revokes provider consent.
Healthcare Provider	Searches authorized patients; views consented histories; records encounters, diagnoses, reports, treatments, and prescriptions.
System Administrator	Manages user accounts, roles, permissions, and system configuration; does not directly access clinical records.
Compliance Auditor	Read-only review, search, and export of immutable audit records; investigates security/compliance events.
External Lab / Hospital System	Exchanges authenticated HL7 FHIR R5 healthcare resources with DHRMS.

Important access rule: System Administrators manage accounts and permissions but must not directly access patient diagnoses, notes, or clinical records. Healthcare Providers can access only records covered by active patient consent.

3. Use Case Specifications

Preconditions describe what must be true before the use case starts. Postconditions describe the resulting state after success or controlled denial. The functions below are based on the functional scope and Appendix B of the SRS.

UC-01 — Authenticate User & Verify MFA

Actor(s): Patient, Healthcare Provider, System Administrator, Compliance Auditor

Precondition: User has a registered, active DHRMS account.

Postcondition: User has an authenticated, time-limited session; successful login is recorded in the audit trail.

Function: Validates credentials (and TOTP when enabled), creates a signed session/JWT, and opens the role-specific dashboard.

UC-02 — View Patient Health Records

Actor(s): Patient; Healthcare Provider

Precondition: User is authenticated; patient has a registered health profile. Provider access additionally requires active, in-scope consent.

Postcondition: Permitted health records are displayed and the read event is logged; otherwise access is denied and the failed attempt is logged.

Function: Retrieves longitudinal medical history, encounters, vitals, and related clinical information after role and consent checks.

UC-03 — Manage Consent

Actor(s): Patient

Precondition: Patient is authenticated in the Patient Portal.

Postcondition: Consent directive is created, updated, revoked, or expired in the database; the change is permanently audited. Revocation blocks future covered provider requests.

Function: Allows the patient to grant, review, narrow/modify, and revoke provider access by data scope and validity duration.

UC-04 — Manage Clinical Encounter

Actor(s): Healthcare Provider

Precondition: Provider is authenticated and authorized for the selected patient; required consent is active.

Postcondition: Validated encounter is stored against the patient and provider; creation or modification is audited.

Function: Records consultation details, timestamp, encounter type, SOAP notes, observations, vitals, and diagnosis.

UC-05 — Manage Diagnostic Reports

Actor(s): Healthcare Provider; Patient (permitted retrieval)

Precondition: Provider is authenticated and authorized for the patient; report is linked to a valid patient/encounter. Retrieval requires applicable consent.

Postcondition: Report metadata/file is saved or permitted results are displayed; operation is audited.

Function: Creates, uploads, retrieves, and manages laboratory/imaging results, findings, test codes, and reference ranges.

UC-06 — Manage Treatment & Prescriptions

Actor(s): Healthcare Provider; Patient (view permitted records)

Precondition: Provider is authenticated and authorized for the patient and encounter; required consent is active.

Postcondition: Treatment/prescription is stored against the correct patient and encounter; creation/modification/query is audited.

Function: Documents treatment plans and creates structured e-prescriptions including medication, dosage, frequency, route, and duration.

UC-07 — Manage Medical Documents

Actor(s): Patient; Healthcare Provider (authorized scope)

Precondition: User is authenticated and authorized for the target patient/document; provider access must satisfy consent scope.

Postcondition: Document metadata and integrity hash are stored with the file in encrypted storage; upload/download/deletion is audited.

Function: Uploads and retrieves permitted PDF/JPEG/PNG/DICOM documents after file-type, size, and SHA-256 integrity validation.

UC-08 — Manage Users, Roles & Permissions

Actor(s): System Administrator

Precondition: Administrator is authenticated with the required administrative privilege.

Postcondition: Account/role/permission/configuration change is saved and audited; administrator cannot directly access clinical records.

Function: Creates, locks, resets, or deactivates accounts; assigns roles and permission sets; manages system configuration.

UC-09 — Review & Export Audit Records

Actor(s): Compliance Auditor

Precondition: Auditor is authenticated and has read-only audit privileges.

Postcondition: Matching audit events are displayed or exported; the review/export action is itself logged where applicable.

Function: Searches and filters immutable audit events by actor, action, resource, date, and status; supports compliance reporting.

UC-10 — Exchange Healthcare Data (FHIR)

Actor(s): External Lab / Hospital System

Precondition: External system is registered/authorized and communicates through an authenticated secure channel (mTLS/OAuth2).

Postcondition: Validated FHIR resources are imported/exported; exchange is recorded in the audit trail and a response is returned.

Function: Exchanges supported Patient, Encounter, Observation, and DiagnosticReport resources using HL7 FHIR R5 over HTTPS.